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A Comparative Volumetric Analysis of the Amygdaloid Complex and Basolateral Division in the Human and Ape Brain

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ABSTRACT The amygdaloid complex functions to facilitate effective appraisal of the social environment and is an essential component of the neural systems subserving social behavior. Despite its critical role in mediating social interaction, the amygdaloid complex has not attracted the same attention as the isocortex in most evolutionary analyses. We performed a comparative analysis of the amygdaloid complex in the hominoids to address the lack of comparative information available for this structure in the hominoid brain. We demarcated the amygdaloid complex and the three nuclei constituting its basolateral division, the lateral, basal, and accessory basal nuclei, in 12 histological series representing all six hominoid species. The volumes obtained for these areas were subjected to allometric analyses to determine whether any species deviated from expected values based on the other hominoids. Differences between groups were addressed using nonparametric comparisons of means. The human lateral nucleus was larger than predicted for an ape of human brain size and occupied the majority of the basolateral division, whereas the basal nucleus was the largest of the basolateral nuclei in all ape species. In orangutans the amygdala and basolateral division were smaller than in the African apes. While the gorilla had a smaller than predicted lateral nucleus, its basal and accessory basal nuclei were larger than predicted. These differences may reflect volumetric changes occurring in interconnected cortical areas, specifically the temporal lobe and orbitofrontal cortex, which also subserve social behavior and cognition, suggesting that this system may be acted upon in hominoid and hominid evolution. Am J Phys Anthropol 134:392–403, 2007.

The evolution of human cognition and its material substrate, the brain, has been a central concern in the study of human evolution. Despite this fact, very few comparative neuroanatomical analyses have focused on hominoid evolution. This study provides volumetric information about an important subcortical structure, the amygdaloid complex (AC) and the nuclei of its basolateral division (BLD), the lateral (L), basal (B), and accessory basal (AB) nuclei, for the often omitted or undersampled hominoids. Because many reports indicate that the AC is integral to social behavior (Brothers and Ring, 1992; Kling and Brothers, 1992; Adolphs, 2003) and that social pressures may have influenced the evolution of the human and primate brain (Dunbar, 2003), this analysis focuses on potential changes in the size and organization of the AC.

Many researchers consider the AC to be a central component of the neural system underlying social cognition and affiliation (Kling, 1986; Brothers, 1990; Adolphs, 1999). Generally, the AC subserves several functions relevant to social behavior like implicit learning, attending to salient stimuli, and memory modulation and consolidation (Phelps, 2005). In humans, AC activation occurs in response to a number of behaviors associated with appraising the social environment, including gaze monitoring (Emery, 2000), processing emotional vocal, facial, and full body expressions (Yang et al., 2002; Hadjikhani and de Gelder, 2003; Glascher et al., 2004; Sander et al., 2005), evaluating another's trustworthiness (Grezes et al., 2004; Singer et al., 2004), and deciding whether to conform to peers' suggestions (Berns et al., 2005). Neurons in the macaque AC respond selectively to dynamic aspects of social behaviors, e.g., social interactions (Brothers and Ring, 1992, 1993) and the social approach of other monkeys (Kling et al., 1979), as well as static representation, e.g., images of faces (Brothers, 1990). Generally, the AC can be seen to facilitate the production of emotional, neural, and bodily responses to various forms of external social stimuli, which serve to direct attention based on the emotional significance of the stimulus and may be important for the rapid assessment of conspecifics in the social environment (Adolphs, 1999).

Such rapid environmental assessment may be a key factor determining an individual's ability to respond to the complexity inherent in the primate social environment, which is suggested to have fostered the evolution of complex cognition (Humphrey, 1988; Whiten and Byrne, 1988; Dunbar, 2003). Many studies have statistically assessed the relationship between the volume of brain components and measures of sociality (Dunbar, 1998), but most of these evolutionary analyses have relied on gross measures like isocortex (neocortex) volume (e.g., Kudo and Dunbar, 2001; Reader, 2003; Byrne and Corp, 2004) despite the fact that not all isocortical components, e.g., the dorsolateral frontal cortex (Raleigh

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and Steklis, 1981), directly participate in the specific network of neural structures thought to subserve social behavior (Brothers, 1990; Adolphs, 1999). Moreover, neuroscientists have stressed the functional links between subcortical limbic structures and isocortex in this network, underscoring the interdependence of emotion and higher order cognition in the production of social behavior (e.g., Damasio, 1994). Thus, to assess this theory from a neuroscientific perspective, more information is needed about both subcortical and cortical structures functionally associated with social behavior.

Limbic structures have traditionally been conceived of as evolutionarily conserved (MacLean, 1990), but considerable evidence points to the fact that limbic structures are more evolutionarily dynamic than previously understood. For example, subcomponents of the orbitofrontal cortex (Semendeferi et al., 1998, 2001) and thalamus (Armstrong, 1986), two limbic structures strongly connected with the AC, show differential volumetric increase across primate species indicating that they are reorganized, an important mechanism of evolutionary change (Holloway, 1973). Stephan and Andy (1977) found that the primate AC appeared to be undergoing evolutionary expansion driven in large part by the "marked development" of, what they termed, the corticobasolateral amygdala (a division of the AC predominantly composed of the BL). In contrast, the remaining portion of the AC (including predominantly the medial and central nuclei) appeared more conserved. This differential expansion indicates that the AC also may be reorganized. Further, although Joffe and Dunbar (1997) found no relationship between measures of sociality and the limited number of anthropoid volumes reported in Stephan and colleagues' (1981) early AC data set (n = 8), Barton and Aggleton (2002) determined that corticobasolateral volume was significantly related to primate social group size using Stephan and colleagues' (1987) later, more comprehensive anthropoid AC data set (n = 27) that included a larger number of hominoids and a more diverse array of primate species overall. Taken together, these studies suggest that the evolutionary reorganization of the primate AC may have been influenced by social factors.

Previous evolutionary analyses of the AC have highlighted the potential importance of functionally related brain regions on the elaboration of amygdaloid subcomponents in primates. The corticobasolateral division's strong interconnections with the expansive isocortex are thought to influence its proliferation, especially given that the constituent nuclei of the more conserved portion of the AC generally target brainstem and olfactory centers (Stephan and Andy, 1977; Barton et al., 2003). Thus, because they communicate with isocortex the BLD nuclei are most likely to show evolutionary change. Yet, the individual BLD nuclei also show different patterns of isocortical connectivity that may influence their development. The L exhibits the most specialized connectivity, predominantly receiving information from the temporal lobe (Stefanacci and Amaral, 2002). The B and AB are more connected to the orbitofrontal cortex, e.g., Brodmann's areas (BA) 12 and 13 (Carmichael and Price, 1995). The AB also receives strong projections from the frontal pole (BA 10) and from the superior temporal sulcus (Stefanacci and Amaral, 2002). Although the role that each nucleus plays in stimulus processing is not certain, it has been hypothesized (Emery and Amaral, 2000; Stefanacci and Amaral, 2002) that the L receives and categorizes sensory information from the temporal cortex, which is then distributed to the B. In the B the processed information is paired with information from the orbitofrontal cortex concerning the social context in which the signal is occurring. After this evaluation, a context appropriate response may be directed through the B's connections with the striatum and the central nucleus (which subsequently targets hypothalamic and brainstem nuclei). Thus, the BLD is in a position to influence a number of functions important for social behavior, e.g., pairing affective values with incoming stimuli, associative learning (Sah et al., 2003), and modulating memory consolidation (McIntyre et al., 2003), which are made possible through the extensive interconnections its constituent nuclei share with specific isocortical areas.

Here we present volumes for the AC (Fig. 1) and the L, B, and AB nuclei (Fig. 2), which form the BLD of the AC. Our study uses a considerably larger sample of hominoids than previous comparative analyses (Stephan and Andy, 1977; Stephan et al., 1987) and includes humans (*Homo sapiens*), all great ape (*Pan troglodytes*, *Pan paniscus*, *Gorilla gorilla*, and *Pongo pygmaeus*), and some lesser ape species (*Hylobates lar, Hylobates concolor*), to investigate the evolution of the AC in this taxon. Because recent analyses indicate that the L is the largest of the human BLD nuclei (Schumann and Amaral, 2005), and this is not true of Old World monkeys (Amaral et al., 1992; Emery et al., 2001), we were particularly interested in determining whether this volumetric increase is a shared hominoid feature or is derived in humans, suggesting reorganization of the AC. Equally, because it subserves social interaction, the proliferation of the AC and its nuclei may be influenced by a species' socioecology; thus, we wanted to examine the potential influence of sociality on the evolution of the AC by addressing differences between the orangutan AC and the AC of the other, more gregarious, great apes.

MATERIALS

Our sample (Table 1) consisted of 12 complete series of histologically processed brains (24 hemispheres) from all six hominoid species including *Homo sapiens* (n = 1), *Pan troglodytes* (n = 3), *Pan paniscus* (n = 2), *Gorilla gorilla* (n = 1), *Pongo pygmaeus* (n = 3), *Hylobates concolor* (n = 1), and *Hylobates lar* (n = 1). For each series, brains were extracted within 12 h after death and submerged in a 4% formalin solution. They were then embedded in paraffin and sectioned in the coronal plane at a thickness of 20 μm. Every 10th to 16th section was Nissl stained using a modification of the Gallyas silver stain for neuronal perikarya (Gallyas, 1971; Merker, 1983). On exception, the chimpanzee YN89-278, was cut in the axial plane at 15 μm thick and every 30th section was stained.

The individuals in our sample were donated by zoological institutes and the Yerkes National Primate Research Center following their natural deaths. Only three older ape specimens (two orangutans, Harry and Briggs, and one chimpanzee, Bathsheba) were wild born. Specimens ranged from 2 to 75 years and represented both sexes (Table 1). Despite this large age range, age related differences in AC volume, independent from whole brain size, are not expected to be large enough to influence interspecific comparisons. Adult humans generally show no significant age related variation in AC volume (Mu

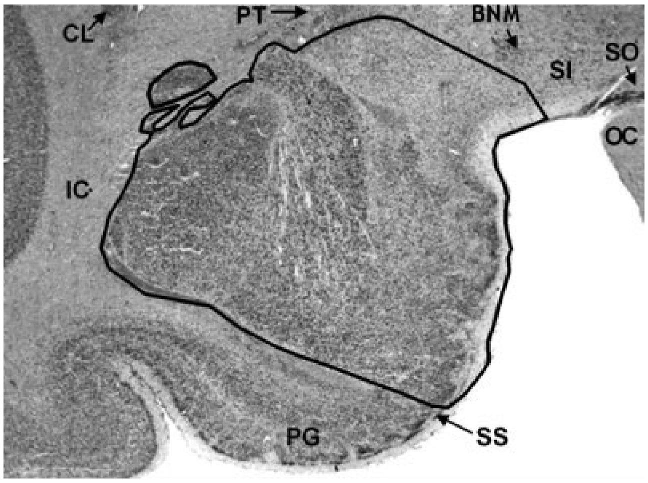

Fig. 1. Gibbon amygdaloid complex in the left hemisphere at the midrostrocaudal level demarcated on a coronal Nissl section. The superimposed lines indicate the anatomical boundaries of the amygdaloid complex at this level. BNM, basal nucleus of Meynert; CL, claustrum; IC, internal capsule; OC, optic chiasm; PG, parahippocampal gyrus; PT, putamen; SI, substantia innominata; SO, supraoptic nucleus; SS, sulcus semiannularis.

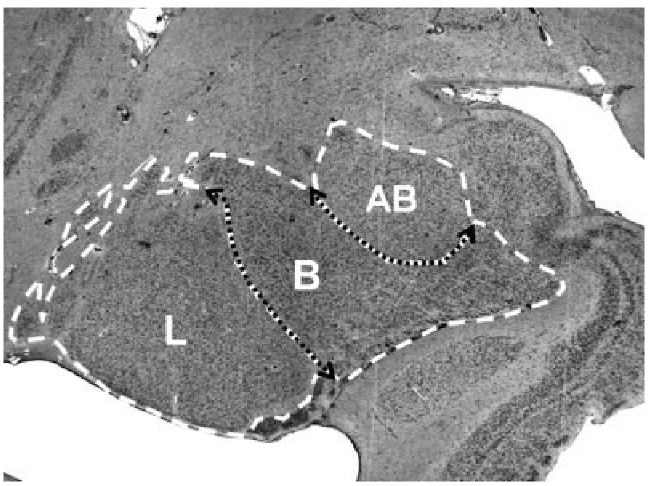

Fig. 2. Human basolateral division in the left hemisphere at the midrostrocaudal level demarcated on a coronal Nissl section. The superimposed white lines represent the anatomical borders of the lateral (L), basal (B), and accessory basal (AB) nuclei. Black dashed lines show the position of the lamina dividing the basolateral nuclei.

et al., 1999; Pruessner et al., 2001). Although one meta-analysis found that adult human AC volumes uncorrected for overall brain volume show a loose negative correlation with age ($\sim r = 0.3/r^2 = 0.09$; Brierley et al., 2002), volumetric differences in the AC generally reflect overall patterns of decrease in gross brain size, gray matter, and neighboring cortical structures (Good et al., 2001; Allen et al., 2005; Walhovd et al., 2005). Thus, the use of relative measures in group comparisons should factor out possible minor differences in adult AC size due to age effects. This is the largest sample of hominoids ever used in a histological comparative study of the AC. Nevertheless, because of specimen availability the sample is necessarily limited and our findings will be treated accordingly.

METHODS

In both hemispheres, the boundaries of the AC, L, B, and AB were hand traced in serial sections by NB with the StereoInvestigator software (MicroBrightField, Williston, VT) using the 1× and 2× objectives (N.A. 0.4 and 0.06) of the Nikon Eclipse E400 microscope.

Volumetric data were collected by NB with the Cavalieri method, using a 150 μm point counting grid in the StereoInvestigator program. In each hemisphere, 9–12 sections were analyzed at intervals of 30–150 μm for each region of interest (ROI), depending on absolute brain size. As required by the Cavalieri method, sampled sections from each specimen were chosen at standard intervals for each ROI and the starting section was picked randomly from the first interval. The coefficient of error (Gundersen, $m = 1$) was less than 0.012 for each specimen; such low values indicate that the precision of the volume estimates was high and that the sampling parameters were sufficient.

We calculated a correction factor (CF) for each specimen to account for much of the shrinkage that occurs during tissue processing. Although CFs cannot account entirely for the possibility that white and gray matter shrink differentially (Kretschmann et al., 1982), procuring a CF for each individual is the best means available to us for nullifying variation due to processing. For each specimen, we divided the preprocessing brain volume by the postprocessing volume, and then multiplied ROIs by this factor. Brain volumes before processing were determined by dividing brain weight by the specific gravity of brain tissue (1.036). Fixed brain volumes were estimated with the Cavalieri method using 50 sections in each series. All CFs were obtained by NB, except for one. We used a CF provided by Carol MacLeod (Langara College, Vancouver) for Schimp because she based her measurement on a complementary series in the C and O Vogt Institute's collection in Düsseldorf, Germany, which included the cerebellum, unlike our series.

Anatomical regions of interest

The AC is a roughly ovoid structure located in the anteromedial temporal lobe (Fig. 1), containing at least 13 distinct nuclei in primates (Amaral et al., 1992). The basolateral (BLD) nuclei (L, B, and AB) are located in the ventrolateral portion of the AC (Fig. 2). The AB is the most medial and dorsal of the BLD nuclei, the L occupies the entire lateral extent of the BLD, and the B lies intermediate between the two (Fig. 2). For this study, definitions of the hominoid AC, L, B, and AB were based on anatomical descriptions of the macaque (Price et al., 1987) and human (de Olmos, 1990; Heimer et al., 1999; Schumann and Amaral, 2005) AC. Because the AC is structurally similar across the primates (Heimer et al., 1999) the boundaries of the ape AC were consistent with these sources. The anatomical criteria for each ROI delineated in this project were as follows.

Amygdaloid complex. The rostral pole of the AC was marked by the appearance of one of the BLD nuclei. In anterior sections the internal capsule borders the AC laterally (Fig. 1). Medially and rostrally, the striated composition of the adjacent piriform cortex contrasts with the diffuse cellular arrangement of the amygdaloid subnuclei. White matter forms the dorsal and ventral borders at this level. In posterior sections, the putamen forms a portion of the dorsal and lateral border, but it is

TABLE 1. Volumetric estimates of the amygdaloid complex and the basolateral division

					Amygdaloid complex[a]		Basolateral[a,c]		Lateral[a]		Basal[a]		Accessory basal[a]	
	Specimen ID	Age	Sex	Hemi[a,b]	L	R	L	R	L	R	L	R	L	R
Human	SN207/84	75	M	1151	2.032	1.773	1.191	1.233	0.551	0.595	0.452	0.429	0.1879	0.2085
Chimpanzee	Bathsheba	24	F	286	0.584	0.561	0.379	0.353	0.126	0.105	0.183	0.187	0.0701	0.0615
	Schimp	22	F	318	0.685	0.733	0.432	0.477	0.138	0.136	0.224	0.255	0.0701	0.0870
	YN89-278[d]	22	M	326	0.634	0.748	–	–	–	–	–	–	–	–
Bonobo	Zahlia	11	F	248	0.623	0.660	0.417	0.445	0.124	0.137	0.223	0.236	0.0695	0.0721
	YN86-137	2	F	304	0.654	0.634	0.425	0.420	0.159	0.164	0.203	0.191	0.0623	0.0660
Gorilla	YN82-140	20	F	300	0.651	0.655	0.442	0.423	0.100	0.100	0.247	0.238	0.0958	0.0855
Orangutan	Harry[e]	37	M	351	0.688	0.660	0.447	0.422	0.121	0.127	–	–	–	–
	Briggs[f]	34	M	279	0.528	0.520	–	0.308	–	0.105	–	0.151	–	0.0516
	YN85-38	16	M	309	0.637	0.514	0.415	0.307	0.124	0.105	0.228	0.153	0.0622	0.0489
Gibbon	Disco	22	F	93.2	0.270	0.367	0.169	0.179	0.060	0.054	0.086	0.086	0.0223	0.0387
	YN81-146	Adult	F	68.3	0.203	0.204	0.130	0.129	0.046	0.041	0.063	0.065	0.0206	0.0236

[a] All volumes are in cm^3.
[b] "Hemi" represents the summed volume of both cerebral hemispheres.
[c] The volume of the basolateral division is equal to the sum of the volumes of the accessory basal, basal, and lateral nuclei.
[d] Specimen is sectioned in the axial plane. Only whole amygdaloid complex volumes for right and left hemispheres were collected to avoid inconsistencies between boundary definitions in the coronal and axial planes for the less distinct nuclei.
[e] The basal and accessory basal nuclei could not be accurately discriminated in this specimen due to localized tissue damage.
[f] Tissue damage in the left temporal lobe prevented collection of lateral, basal, and accessory basal nuclei in the left hemisphere of this specimen. Volumes in the right hemisphere were doubled in this specimen in the analysis.

distinguishable from the AC based on differences in cell structure, density, and organization. The AC is also bounded dorsally by the substantia innominata, marked by the presence of the basal nucleus of Meynert in this analysis (Fig. 1). In caudal sections, the sulcus semiannularis (Fig. 1) separates the entorhinal cortex from the AC medially (Schumann and Amaral, 2005), while the lateral ventricle and hippocampus form the dorsal borders.

Lateral. Because the L is the most lateral of all of the nuclei, its lateral, dorsal, and ventral borders are consistent with those of the AC. Rostrally, the L is in close proximity to the claustrum, but the latter structure is distinguishable by its larger, more darkly staining cells. The medial border is defined by the presence of the lateral medullary lamina (Fig. 2) and the fact that the cells of the L are smaller and more compact than those in the neighboring B. Caudally, the comparatively larger cells of the dorsal part of the L distinguish it from the adjacent putamen.

Basal. The B is separated from other nuclei by the longitudinal association fiber bundle. The lateral medullary lamina divides the lateral aspect of the B from the L, while the intermediate medullary lamina divides the medial aspect of this division from the AB (Fig. 2). The dorsomedial border separating the B and AB is best discerned by changes in cell size.

Accessory basal. The intermediate medullary lamina marks the lateral border of the AB as does the absence of the larger cells that characterize the neighboring B. The medial border is demarcated by the medial medullary lamina, which divides the AB from the superficial cortical nuclei. The ventral border is the same as that for the dorsomedial border of the B.

Data analysis

Absolute volume measures for each ROI discussed in the text include the summed volumes of the structure in the right and left hemisphere. The relative volume of the ROI is this absolute volume divided by the total volume of both hemispheres, in other words, the percent of the hemispheres occupied by the ROI. The cerebral hemispheres ("hemispheres") were defined as the whole brain minus cerebellum, midbrain, and brainstem. To assess the possibility of volumetric lateralization, the relative degree of asymmetry was measured as: $|(\text{Left ROI} - \text{Right ROI})/[(\text{Left ROI} + \text{Right ROI})/2]|$.

Regression analyses were performed using log-transformed data. Equations and 95% confidence intervals from standard least squares regression (SLS) were obtained using JMP IN 4.04 (SAS Institute, 2000) statistical software. In previous studies (Semendeferi and Damasio, 2000) we found minor differences between SLS and major axis regressions, so here we used only the more common SLS. We also performed independent contrasts (IC) analysis with the PDAP program (Garland et al., 1999; Garland and Ives, 2000), using Purvis's (1995) phylogeny to set branch lengths (which were squared to ensure that standardized contrasts did not correlate with their standard deviation). Regressions drawn from the PDTREE module of PDAP were mapped back into original data space with individual values for each specimen plotted over this regression line. For the independent variable in all regressions, ROI volume was subtracted from hemisphere volume to avoid statistical artifacts related to regressing an ROI against a region of which it is a part (Deacon, 1990). We report results from both regressions focusing on the allometric relationship between ROI volume and hemisphere volume in two conditions: including or excluding the human datum. In all cases, SLS regressions yielded considerably narrower 95% confidence intervals (CI) than IC analysis. Differences between slopes were tested using a modified version of the *t*-test (Zar, 1996: section 18.1). Because this sample is unlikely to satisfy the requirements of parametric analyses, e.g., normal distribution, the nonparametric Wilcoxon two-sample rank test was used to test for differences between means in JMP IN 4.04.

RESULTS

Absolute volumes

Amygdala and basolateral division. In absolute volume, the human set the upper bound for the hominoid

AC and BLD at 3.805 cm^3 and 2.424 $cm,^3$ respectively, while the gibbons set the lower bound, ranging from 0.407 to 0.637 cm^3 (AC) and 0.259 to 0.348 cm^3 (BLD) (Table 1). Great ape AC volumes ranged from 1.048 cm^3 in an orangutan to 1.418 cm^3 in a chimpanzee. Also, a chimpanzee had the largest (0.909 cm^3), and an orangutan had the smallest (0.616 cm^3) absolute BLD volume. All great ape species overlapped for both ROIs (Table 1).

Basolateral Nuclei (L, B, AB). The human exhibited the greatest absolute volumes for the L (1.146 cm^3), B (0.881 cm^3), and AB (0.3964 cm^3), and the gibbons exhibited the smallest L (0.087–0.114 cm^3), B (0.128–0.172 cm^3), and AB (0.0442–0.0610 cm^3) (Table 1).

In the great apes, volumes for the L fell between those set by a bonobo (0.323 cm^3) at the high end and a gorilla (0.200 cm^3) at the low end. The values for the B overlapped with the gorilla (0.485 cm^3) setting the upper limit and an orangutan (0.302 cm^3) setting the lower one. AB volume was distinctly largest in the gorilla (0.1813 cm^3) and smallest in the orangutans (0.1032–0.1111 cm^3), while the bonobo and chimpanzee values (0.1283–0.1571 cm^3) were intermediate and overlapped one another.

As expected based on differences in overall brain size, absolute AC volumes were greatest in the human, smaller and overlapping in the great apes, and smallest in the gibbons. In the great apes, the BLD and its constituent nuclei followed this pattern, with the exception of the AB, which did not show overlap in all great ape species. It was distinctly largest in the gorilla and smallest in the orangutans.

Regression against hemisphere volume

Amygdala. AC volumes were negatively allometric with hemispheres (SLS slope = 0.737 SE = 0.0485; IC slope = 0.755 SE = 0.0735; slope difference: n.s., P = 0.439). The AC of two of the three orangutans fell below the CI and the third fell on its lower periphery (Fig. 3a). Excluding the human from the analysis resulted in a further decrease in slopes (SLS slope = 0.686 SE = 0.0469; IC slope = 0.582 SE = 0.140; slope difference: n.s., P = 0.0928).

Basolateral division. The volume of the BLD was also negatively allometric with the hemispheres (SLS slope = 0.769 SE = 0.053; IC slope = 0.749 SE = 0.093; slope difference: n.s., P = 0.642). The BLD of two orangutans fell below the CI, and all three fell below the regression line (Fig. 3b). When the human BLD was excluded, the slope changed slightly in the SLS analysis (slope = 0.736 SE = 0.0548) and clearly decreased in the IC analysis (slope = 0.594 SE = 0.258; slope difference: n.s., P = 0.229*).* The rest of the AC (AC minus BLD) and the hemispheres were negatively allometric with the human included (SLS slope = 0.769 SE = 0.0623; IC slope = 0.784 SE = 0.0430; slope difference: n.s., P = 0.799) and with the human excluded (SLS slope = 0.603 SE = 0.0256; IC slope = 0.585 SE = 0.0352; slope difference, n.s., P = 0.904).

Lateral nucleus. The relationship between the L and the hemispheres (Fig. 4a) was the closest to isometric of all the analyses in which all species were included (SLS slope = 0.852 SE = 0.090; IC slope = 0.991 SE = 0.216; slope difference: n.s., P = 0.392). When the human value was excluded (Fig. 4b), the relationship between the L and the hemispheres became more negatively allometric (SLS slope = 0.674 SE =0.066; IC slope = 0.361 SE = 0.508; slope difference: n.s., P = 0.666). Unlike any other

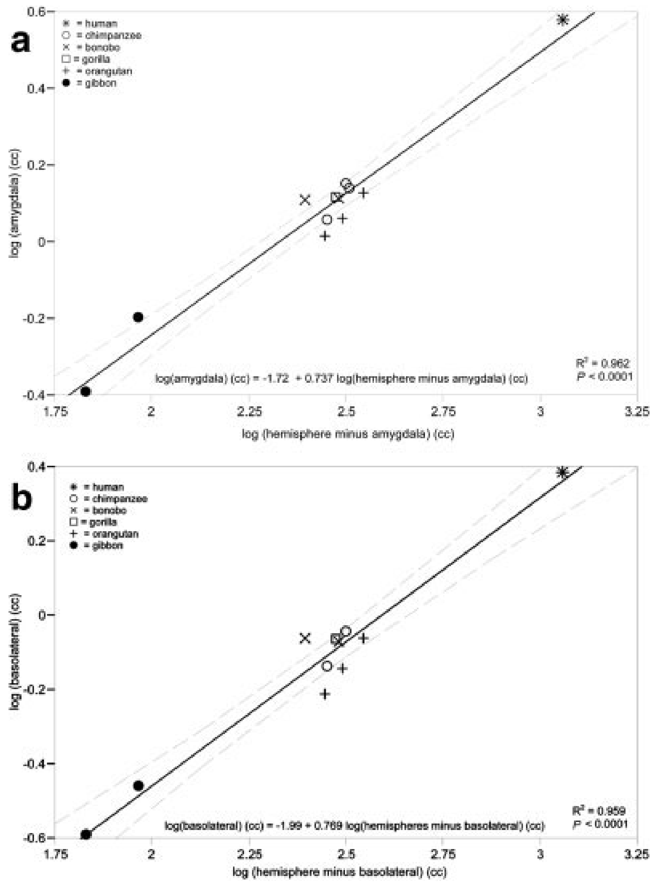

Fig. 3. Least squares regression plotting the log of hemisphere volumes against the log of the volumes of (**a**) the amygdala (amygdaloid complex) and (**b**) the basolateral division with all species of hominoid included in the regression.

region analyzed, the value for the human L fell above the regression line and the CI in this analysis.

Basal nucleus. The relationship between the B and the hemispheres was the most negatively allometric of all tested combinations when all species were included (SLS slope = 0.689 SE = 0.073; IC slope = 0.550 SE = 0.083; slope difference: n.s., P = 0.473). The gorilla's B fell above the confidence interval in this analysis (Fig. 4c). Without the human datum, the slope of the regression line increased (SLS slope = 0.770 SE = 0.071; IC slope = 0.680 SE = 0.236; slope difference: n.s., P = 0.979), indicating that the human value may be driving the slope downward.

Accessory basal nucleus. The relationship between the AB and the hemispheres was negatively allometric with all species included (SLS slope = 0.754 SE = 0.075; IC slope = 0.733 SE = 0.085; slope difference: n.s., P = 0.958). The AB of both analyzed orangutans fell below the confidence interval and the gorilla fell above it (Fig. 4d). Excluding the human resulted in only a minor change in slope (SLS slope = 0.732 SE = 0.079; IC slope = 0.770 SE = 0.254; slope difference: n.s., P = 0.973).

The human L was significantly larger than predicted (Fig. 4b). Also, in the analysis of the B, including the human appeared to negatively influence the slope. In all of the analyses, the orangutan data fell below the regression line, and they fell significantly below the confidence interval for the BLD, the L, and the AB (Figs. 3 and 4).

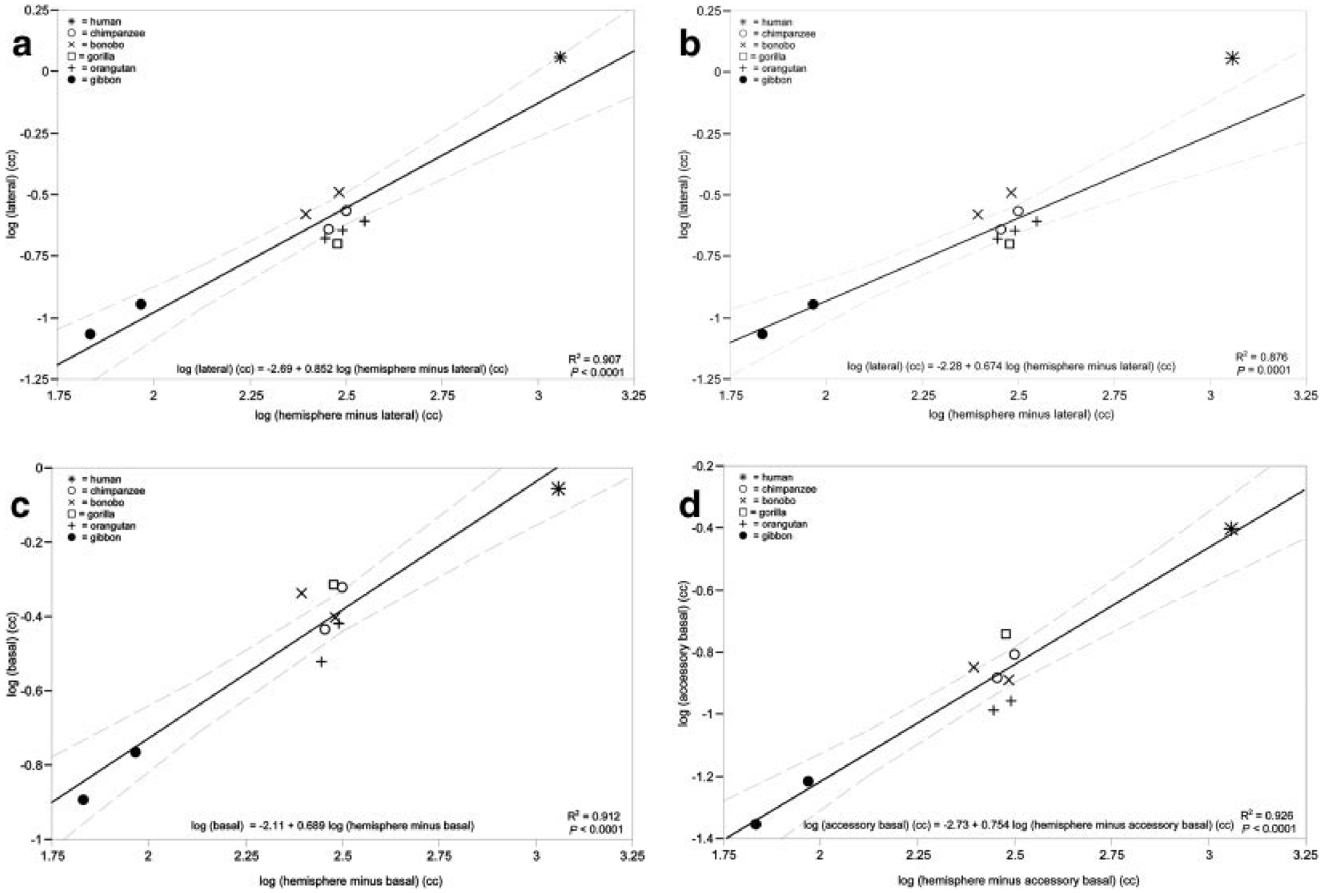

Fig. 4. Least squares regression plotting the log of hemisphere volumes against the log of the volumes of (**a**) the lateral nucleus with all species included, (**b**) the lateral nucleus with the human excluded from the regression, (**c**) the basal nucleus with all species included, and (**d**) the accessory basal nucleus with all species included. The results of additional regression analyses are discussed in the text.

While the gorilla possessed a larger B than predicted, its L fell below the confidence interval (Fig. 4a). Only in these three species did all members fall outside the confidence interval. Several other species did exhibit interesting intraspecific variation with at least one individual falling outside of the range of predicted values in regressions that included all species, e.g., the two bonobos both fell outside the confidence interval for several analyses, but never jointly on the same measure. Unfortunately, the small sample size limits our ability to draw any strong conclusions about this variation. For example, the bonobo which falls above the line for the lateral nucleus (Fig. 4a) was both young and language trained, which may influence her position on the line, but we do not have a suitable comparative sample of language trained or juvenile primates to test these assumptions. Equally, it must be noted that the sample includes only one gorilla and one human and that a larger sample might yield a greater range of volumes due to individual variation. Our findings for the human are consistent with other volumetric studies (see discussion), but few data are available for the gorilla AC and further investigation is needed in this species.

Relative volumes

Amygdala relative to hemispheres. The AC occupied the smallest percentage of the hemispheres in the human (0.33%), the largest in the gibbons (0.60%–0.68%), and overlapped in the African apes (0.40%–0.52%) (Fig. 5a). The mean percent occupied by the orangutan AC (0.37%–0.38%) was significantly smaller than that of the average African ape ($Z = -2.09$; $P = 0.0364$).

Basolateral division relative to hemispheres. The BLD also occupied the smallest percentage of the hemispheres in the human (0.21%) and the largest in the gibbons (0.37%–0.38%). Similar to the AC, the percentage of the hemispheres occupied by the BLD (Fig. 5b) was smaller in the orangutans (0.22%–0.25%) than in the African great apes (0.26%–0.35%) and the difference between the two group means was significant ($Z = -2.10$; $P = 0.0358$). In contrast, in all of the great apes the relative volumes of the rest of the AC (all non-BLD nuclei) fell within a similar range (Fig. 5c) and the mean value of this region was not significantly smaller in orangutans ($Z = -1.19$; $P = 0.233$) when compared with the African apes.

Basolateral division relative to amygdala volume. In all hominoids, the BLD occupied a majority of the AC, from 55 to 67% for most animals with overlap across all species.

Lateral, basal, and accessory basal nuclei volume relative to basolateral volume. The organization of

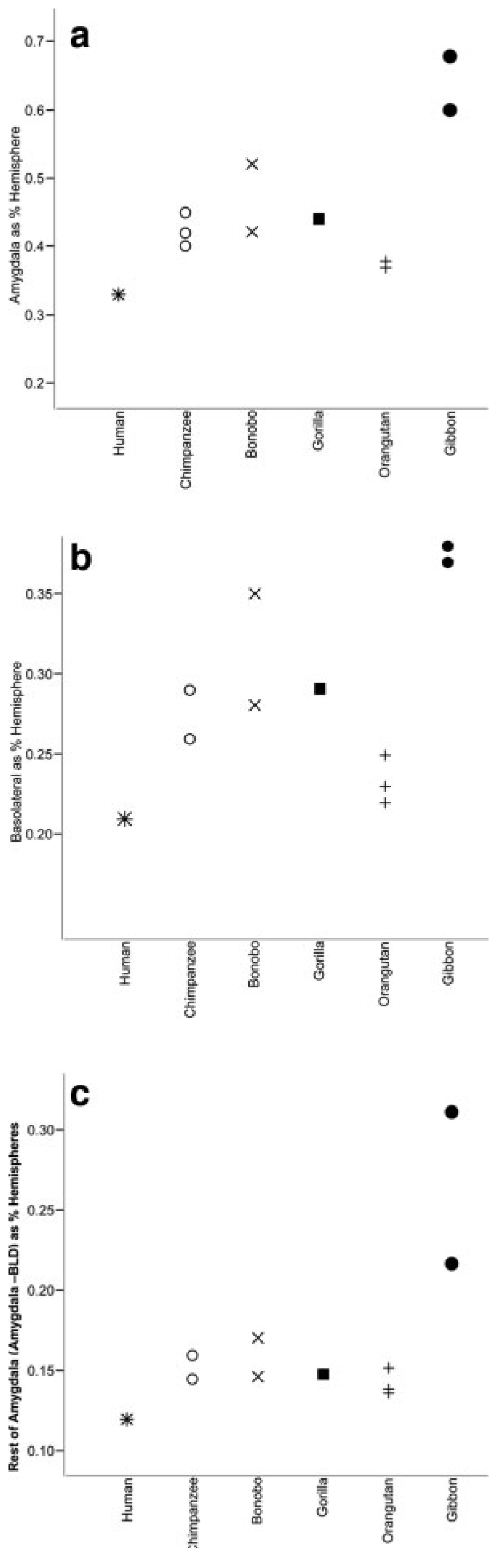

Fig. 5. Comparison of relative volumes (percentage of hemisphere volumes) of (**a**) the amygdala (amygdaloid complex), (**b**) the basolateral division, and (**c**) the rest of the amygdala (amygdaloid complex minus basolateral division) in the hominoids.

the human BLD was distinct from that of the apes. The L occupied the largest percentage of the human BLD, 47%, while the B occupied 36%, and the AB occupied 16%. In contrast, in the apes the B occupied the largest percentage of the BLD followed in succession by the L and the AB (Fig. 6). In the gibbons, orangutans, bonobos, and chimpanzees, the B occupied 46–53%, the L occupied 29–38%, and the AB occupied 15–18% of the BLD. The gorilla is the only species in which the L and AB occupy similar percentages within BLD; the percentages occupied by the B and AB (56% and 21%, respectively) fell above the range of values for the other apes and percentage occupied by the L (23%) fell below this range.

The orangutans had significantly smaller relative AC and BLD volumes than did the African apes. In all species, the BLD occupied the majority of the AC. When the volumes of the individual BLD nuclei were compared relative to BLD volume, the human L appeared exceptional in that it forms the largest component of the BLD, while the B was the largest in the apes. Concordant with the regression analysis, the gorilla had the smallest relative L, but a large AB and B.

DISCUSSION

Our values for the AC in the two human hemispheres fall within the range published in a recent meta-analysis of 39 papers (Brierley et al., 2002) using MR images of living humans (Table 2). Our hominoid data largely overlap with previously reported postmortem human, chimpanzee, orangutan, and gibbon AC volumes procured by Stephan and colleagues (1977, 1987) and Zilles and Rehkamper (1988). The only previously reported gorilla AC volume (Stephan et al., 1987) was more than twice as large as the AC of the gorilla in the present study (Table 2). Although sexual dimorphism may account for the differences between the two studies (the Stephan gorilla was male, while the one in this analysis was female), more specimens are needed to investigate this possible source of difference. The only other postmortem histological analysis of the AC (Schumann and Amaral, 2005) reported volumes uncorrected for shrinkage and thus cannot be directly compared to either our values or the MR studies above (as would be expected, their uncorrected values fall at the lower end of that range; Table 2).

Humans. The L has been suggested to be large in humans (Stephan et al., 1987) and is the largest nucleus in the human BLD (Schumann and Amaral, 2005). In contrast, the B is the largest nucleus in the macaque BLD (Amaral et al., 1992; control data in Emery et al., 2001). In this analysis, we report that the human AC, specifically the BLD, shows a unique pattern of organization which also deviates from that of our closest phylogenetic relatives, the great apes (Fig. 6). Our data suggest that the organization of the human BLD is distinguished by a volumetric increase in the L and also, potentially, a decrease in the B.

The unique organization of the human BLD, in which the L occupies most of the BLD, may reflect the L's primary connective relationship with the temporal lobe (Stefanacci and Amaral, 2002). In addition to the large L, humans appear to have relatively larger temporal lobes (Semendeferi and Damasio, 2000; Rilling and Seligman, 2002) and more temporal white matter (Semendeferi and Schenker, 2001; Rilling and Seligman, 2002) than other apes. Although increases in temporal lobe white matter most likely reflect increased connectivity intrinsic to this region (Schenker et al., 2005), the

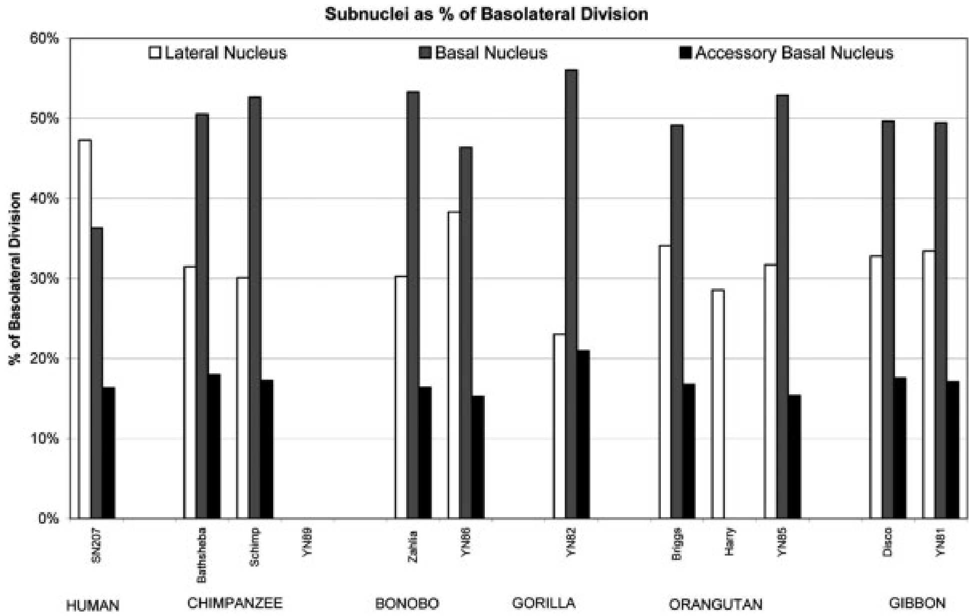

Fig. 6. Percentage of basolateral division occupied by the lateral, basal, and accessory basal nuclei in each specimen.

TABLE 2. Hominoid amygdaloid complex volumes reported in the anatomical literature

Species	Range of amygdaloid complex volumes (cm^3)				
	Present study	Stephan et al.[a]	Zilles and Rehkamper[b]	Brierley et al.[c,d]	Schumann and Amaral[d,e,f]
Human	3.80	3.02 and 5.29	–	2.10–7.76	2.40–3.18
Chimpanzee	1.15–1.42	1.42	–	–	–
Bonobo	1.28 and 1.29	–	–	–	–
Gorilla	1.31	2.75	–	–	–
Orangutan	1.0–1.35	–	1.4	–	–
Gibbon	0.41 and 0.64	0.67	–	–	–

[a] Histological volumes for three ape specimens and one human taken from Stephan, Frahm, and Baron (1987). A different human volume was also presented in Stephan and Andy (1977).
[b] Histological volume from Zilles and Rehkamper (1988).
[c] MR volumes from Brierley et al. (2002), a meta-analysis of 39 studies with 51 human datasets.
[d] The original volumes reported in these studies were from only one hemisphere. To provide a better comparison with other studies in the table, these reported volumes were doubled before adding them to the table.
[e] Histological volumes from Schumann and Amaral (2005). Sample included 10 human hemispheres.
[f] Information from Schumann and Amaral (2005) is not directly comparable to the rest of the table as the study reports post-processing volume and did not use corrections factors to estimate preprocessing volume

temporal lobe is an important component of the system proposed to underlie social cognition (Adolphs, 2003; Brothers, 1990) and overall volumetric increases may have a functional, evolutionary significance. Indeed, Rilling and Seligman (2002) suggest that the elaboration of the human temporal lobe is likely to reflect the importance of language processing in this region. Stephan and Andy (1977) suggested that portions of the AC which include the BLD show greater expansion than other portions of the AC due to the influence of the BLD's strong isocortical connections. Thus, it is likely that the L had expanded, and consequently the BLD is reorganized, in humans at least in part due to evolutionary changes in the temporal lobe and the strong relationship shared between this structure and the L.

The human exhibits the smallest proportion of AC and BLD relative to hemispheres, but shows a positive residual for the latter value. It is interesting that, while the AC residual was not positive, the BLD residual was. This may reflect increased processing demands within the BLD due to a preponderance of information flowing into the human AC, or it may even indicate greater cortical control over the AC. However, the residual was not significantly positive, so any conclusions on this issue require increased samples.

Orangutans. Our results suggest that orangutans are unique among the apes in having a smaller AC, possibly driven by their smaller BLD. Within the BLD we found that the AB and B nuclei were smaller in orangutans than in the African apes. The BLD and especially AB receive considerable projections from orbitofrontal cortices (Ghashghaei and Barbas, 2002; Stefanacci and Amaral, 2002), which are also smaller in orangutans (Semendeferi et al., 1997; Schenker et al., 2005). Further, the organization of BA 13, a major constituent of the orangutan orbitofrontal cortex, appears more "prefrontal" than "limbic" (Semendeferi et al., 1998). Thus, it is likely that specializations are present in the orangutan brain that de-emphasize limbic structures.

Behavioral evidence accords with this possible neural difference in orangutan limbic structures, which generally mediate affect as well as social interactions. While few comparative analyses remark upon affective differences between ape species, Shumaker et al. (2001) found that orangutans appeared to engage in less impulsive decision making than chimpanzees when presented with edible stimuli in a numerical ordination task. The authors postulated that this may be evolutionarily related to the reduced amount of direct competition for food in the orangutan social environment. Wild orangutans are unique among the hominoids as they are the only semi-solitary species, occasionally forming bands averaging 2 individuals (Delgado and van Schaik, 2000). In contrast, members of the highly gregarious African genus *Pan* live in multimale-multifemale groups of up to 150 individuals (bonobos: Kano, 1992; chimpanzees: Mitani and Amsler, 2003), and western lowland gorillas typically form single male groups of up to 20 individuals (Yamagiwa et al., 2003). Given this difference in socioecology, the finding that orangutans show both emotional and neural differences compared with other apes is highly provocative. Unfortunately, our certainty as to whether these results reflect real population level differences that are completely genetic in origin is limited by the small samples available and the diverse rearing conditions experienced by the apes in them. Future studies with larger samples are needed to confirm these ideas.

Gorillas. The gorilla's BLD comprised an exceptionally large B and AB and a diminished L, a more extreme manifestation of the pattern found among the other apes. The fact that the gorilla had a particularly small L corresponds with a report indicating that gorillas have the smallest temporal lobe among the apes (Semendeferi and Damasio, 2000; Rilling and Seligman, 2002), representing a reversal of the pattern found in humans (large L and large temporal lobe). One could argue that this supports the idea that changes in isocortical volume inspire concomitant changes in related AC nuclei volume. Accordingly, one would expect that the large size of the gorilla AB would correspond to an expansion in interconnected isocortical regions. This is not patently the case. While the AB shares strong connections with the superior temporal gyrus and the orbitofrontal cortex (Stefanacci and Amaral, 2002), gorillas do not have larger superior temporal gyri (Rilling and Seligman, 2002) and only a slightly larger percentage of the frontal lobes is devoted to the orbitofrontal cortex in gorillas than in the other apes (Semendeferi et al., 1997). Still, they do not have appreciably smaller superior temporal gyri than the other apes, despite having relatively smaller temporal lobes (Rilling and Seligman, 2002; Semendeferi and Damasio, 2000). Also, BA 13 of the gorilla's oribitofrontal cortex, while not larger than in the other apes, is more cell dense and contains higher numbers of neurons (Semendeferi et al., 1998). Thus, it is possible that the processing capacity and extrinsic connectivity of portions of the gorilla's orbitofrontal cortex are higher (Semendeferi et al., 1998). As such, it may be that changes in the gorilla orbitofrontal cortex drive changes in the AB, which explains why it would not be affected by the lack of extreme volumetric increase in the superior temporal gyrus.

While theories of connectivity can be invoked to explain volumetric differences in the hominoid AC, the difficulty of this claim, overall, is the fact that the some

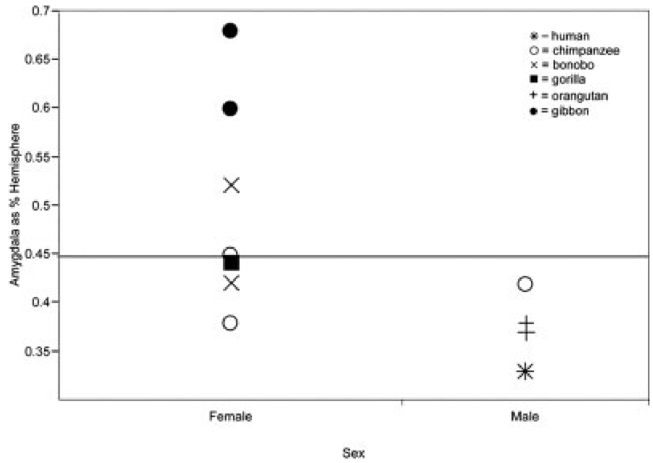

Fig. 7. Range of relative volumes in male and female specimens. Amygdala (amygdaloid complex) volumes are reported as a % of the hemisphere volumes.

nuclei share sparser connections with an array of isocortical areas. The influence of these connections must also be considered in our explanatory framework. While the L is more selectively connected to the temporal lobe, the B receives light projections from the temporal lobe (Ghashagaei and Barbas, 2002) in addition to the strong connections it receives from orbitofrontal cortex. It projects extensively to frontal, temporal, and occipital cortices (Amaral et al., 1992, 2003), making it more difficult to identify any one region that might influence the B's evolution. Although the AB receives strong connections from the orbitofrontal cortex, it is also heavily innervated by some areas within the superior temporal gyrus (Stefanacci and Amaral, 2002). Because our findings are more focused on the AB and L and are consonant with the general connective relationships of these two nuclei, it is very likely that isocortical regions are influencing this hominoid AC development.

Other considerations. Although we have focused on volumetric differences across species, other factors might also influence a structure's size. A short discussion of our findings regarding sex and hemispheric differences is warranted, keeping in mind that none of the results were reliably significant, possibly due, in part, to the size of our sample.

Most species in our sample include members of only one sex (Table 1), a fact which precludes thorough assessment of the effects of sex on AC volume in individual species. Nonetheless, we investigated sex differences across AC volumes in the sample to test the possibility that some of our findings, especially the small orangutan AC, may be the result of variation due to sex and not species. Females in the sample had relatively larger AC than males as a group (Fig. 7). Yet, in the chimpanzees, the only species in which both sexes were sampled, the relative volume of the male's AC fell within the range of the females' volumes, suggesting that this sex difference may be an artifact of sex bias in the sample. This is not surprising, given that there is no concrete evidence for sex differences in the human AC literature. AC volumes (corrected for brain volume) are reported to be larger in males (Goldstein et al., 2001), in females (Szabo et al., 2003), similar between the sexes (Gür et al., 1996; Giedd et al., 2002), or larger in only one hemisphere in males

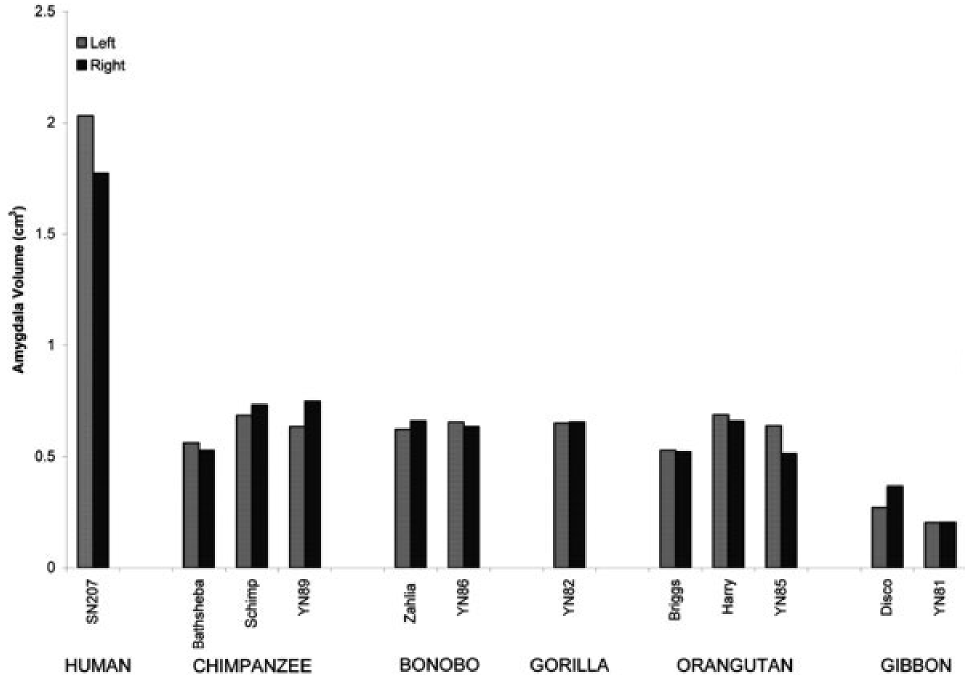

Fig. 8. Graph of absolute right and left amygdala volume in cm^3.

(Brierley et al., 2002). No clear sex difference in AC volume is apparent in chimpanzee (Freeman et al., 2004) or macaque samples either (Franklin et al., 2000). This casts doubt on the presence of sex differences in the sample and also provides partial support for our hypothesis that orangutan AC volumes are distinct due to phylogenetic and ecological influences and not the composition of the sample.

No consistent direction of asymmetry was present in the hominoid AC (Fig. 8) or BLD nuclei, nor did these regions exhibit a large degree of asymmetry across the specimens. Only two specimens, an orangutan (YN85-38) and a gibbon (Disco), showed a degree of asymmetry that exceeded 0.26, i.e., the volume in one hemisphere was over 1.5 times the volume of the other, and this was only for one nucleus in each specimen. These results are consistent with MR analyses using larger human (Brierley et al. 2002) and chimpanzee (Freeman et al., 2004) samples which found no asymmetries in the AC.

CONCLUSIONS

Our findings largely support hypotheses of AC evolution that highlight the importance of functional networks within the brain, specifically those related to social behavior. Although this is the most expansive comparative data set for the hominoid AC to date, the sample is small and all conclusions are, thus, somewhat contingent. The results of this analysis suggest several profitable directions for future research. The species differences we found in the hominoid AC may be related to the connections that the AC shares with other portions of the brain. The large size of the human L may reflect the proliferation of the temporal lobe over the course of hominid evolution, while the inverse may be true of the gorilla. The smaller size of the orangutan AC and BLD may be related to the diminished importance of interconnected limbic structures in this species. Further, there is some evidence that the orangutan, which exhibits one of the smallest group sizes on the continuum of primate sociality, may also be distinguished neuroanatomically from the other great apes, suggesting that social pressures may play a role in the development of the AC in association with other limbic regions. It is beyond the scope of this analysis to test specific measures of social complexity, because such an investigation would require a much larger sample. Given our preliminary findings, though, each of these hypotheses is viable and certainly not mutually exclusive, providing a solid foundation for future investigations of the hominoid AC.

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